

INS Statistical Survey Collection Platform

Database Specification Document

Version 2.0 (Updated After Supervisor Review)

1. Introduction

This document describes the logical database architecture of the INS Statistical Survey Collection Platform.

The objective of this database is to support the complete lifecycle of statistical surveys conducted by the Institut National de la Statistique (INS), including:

- User management
- Role management
- Authentication
- Company management
- Contact management
- Survey management
- Passage management
- Sample management
- Collection monitoring
- Reporting
- Notification management
- Audit tracking

The architecture is designed to be modular, secure, scalable and maintainable.

2. Database Architecture Overview

The database is organized into the following domains:

```
Administration
├─ Roles
├─ Users
├─ System Settings
├─ Audit Logs

Authentication
├─ OTP Codes
```

└─ Login Attempts

Organization

└─ Companies

└─ Contacts

└─ Sectors

└─ Activities

Survey Management

└─ Surveys

└─ Survey Assignments

└─ Periodicities

└─ Survey Passages

└─ Survey Questionnaires

└─ Samples

└─ Sample Companies

└─ Survey Activations

Collection

└─ Collection Tracking

Communication

└─ Notifications

Results

└─ Survey Result Files

3. Roles

Purpose

Stores platform roles.

Table

roles

Fields

Field	Type	Description
id	BIGINT PK	Unique identifier

Field	Type	Description
code	VARCHAR(50) UNIQUE	Role code
name	VARCHAR(100)	Role name
description	TEXT	Description
created_at	DATETIME	Creation date

Initial Roles

SYSTEM_ADMIN

SURVEY_ADMIN

ACCOUNT_MANAGER

COMPANY_CONTACT

4. Users

Purpose

Stores all platform accounts.

Table

users

Fields

Field	Type
id	BIGINT PK
role_id	BIGINT FK
email	VARCHAR(255) UNIQUE
password_hash	VARCHAR(255)
phone_number	VARCHAR(30)

Field	Type
email_verified	BOOLEAN
email_verified_at	DATETIME NULL
status	ENUM('ACTIVE','LOCKED','DISABLED')
first_login	BOOLEAN
preferred_language	ENUM('ar','fr','en')
password_only_until	DATETIME NULL
last_login_at	DATETIME NULL
created_at	DATETIME
updated_at	DATETIME

Business Rules

- Users are never deleted.
 - Email must be unique.
 - Passwords stored using bcrypt.
 - Accounts disabled through status changes.
-

5. Companies

Purpose

Stores participating companies.

Table

companies

Fields

Field	Type
id	BIGINT PK
identifier	VARCHAR(100) UNIQUE
company_name	VARCHAR(255)

Field	Type
tax_number	VARCHAR(100)
sector_id	BIGINT FK
activity_id	BIGINT FK
address	TEXT
governorate	VARCHAR(100)
postal_code	VARCHAR(20)
phone	VARCHAR(30)
email	VARCHAR(255)
status	ENUM('ACTIVE','INACTIVE')
created_at	DATETIME
updated_at	DATETIME

Business Rules

- Companies are never deleted.
 - Identifiers must be unique.
-

6. Contacts

Purpose

Stores company representatives.

Table

contacts

Fields

Field	Type
id	BIGINT PK
user_id	BIGINT FK

Field	Type
company_id	BIGINT FK
first_name	VARCHAR(100)
last_name	VARCHAR(100)
position	VARCHAR(255)
is_primary_contact	BOOLEAN
status	ENUM('ACTIVE','INACTIVE')
created_at	DATETIME
updated_at	DATETIME

Business Rules

- One company may have multiple contacts.
- All active contacts may access surveys assigned to their company.

7. Sectors

Purpose

Reference table for business sectors.

Table

sectors

Fields

Field	Type
id	BIGINT PK
code	VARCHAR(50)
name_fr	VARCHAR(255)
name_ar	VARCHAR(255)
name_en	VARCHAR(255)

8. Activities

Purpose

Reference table for economic activities.

Table

activities

Fields

Field	Type
id	BIGINT PK
sector_id	BIGINT FK
code	VARCHAR(50)
name_fr	VARCHAR(255)
name_ar	VARCHAR(255)
name_en	VARCHAR(255)

9. Periodicities

Purpose

Stores survey frequencies.

Table

periodicities

Fields

Field	Type
id	BIGINT PK
code	VARCHAR(50)

Field	Type
name	VARCHAR(100)

Values

MONTHLY

QUARTERLY

SEMESTRIAL

ANNUAL

BIENNIAL

CUSTOM

10. Surveys

Purpose

Stores INS surveys.

Table

surveys

Fields

Field	Type
id	BIGINT PK
code	VARCHAR(50) UNIQUE
name	VARCHAR(255)
description	TEXT
periodicity_id	BIGINT FK
status	ENUM('ACTIVE','INACTIVE','ARCHIVED')

Field	Type
created_by	BIGINT FK
created_at	DATETIME
updated_at	DATETIME

Business Rules

- Survey codes must be unique.
 - Surveys are never deleted.
 - Archived surveys remain available for consultation.
-

11. Survey Assignments

Purpose

Assigns surveys to Survey Administrators.

Table

survey_assignments

Fields

Field	Type
id	BIGINT PK
survey_id	BIGINT FK
user_id	BIGINT FK
assigned_at	DATETIME

Business Rules

A Survey Administrator may manage only surveys assigned to him.

12. Survey Passages

Purpose

Represents collection campaigns.

Table

survey_passages

Fields

Field	Type
id	BIGINT PK
survey_id	BIGINT FK
year	INT
passage_number	INT
opening_date	DATETIME
closing_date	DATETIME
status	ENUM('DRAFT','READY','OPEN','CLOSED','ARCHIVED')
activated_by	BIGINT FK NULL
activated_at	DATETIME NULL
closed_by	BIGINT FK NULL
closed_at	DATETIME NULL
created_at	DATETIME
updated_at	DATETIME

Constraints

Unique:

survey_id + year + passage_number

Business Rules

- Closing date must be greater than opening date.
 - Passage may be activated only when questionnaire and sample exist.
-

13. Survey Questionnaires

Purpose

Stores external questionnaire references.

Table

survey_questionnaires

Fields

Field	Type
id	BIGINT PK
passage_id	BIGINT FK
questionnaire_url	TEXT
questionnaire_pdf_path	TEXT
version	VARCHAR(50)
status	ENUM('ACTIVE','INACTIVE','ARCHIVED')
created_at	DATETIME

Important Note

Questionnaires are currently hosted on an external platform.

The synchronization mechanism between the questionnaire platform and INS platform is not yet defined.

The following items remain under clarification:

- Questionnaire opening tracking
- Questionnaire completion tracking
- Questionnaire response synchronization

14. Samples

Purpose

Stores survey samples.

Table

samples

Fields

Field	Type
id	BIGINT PK
passage_id	BIGINT FK
uploaded_by	BIGINT FK
total_companies	INT
created_at	DATETIME

15. Sample Companies

Purpose

Stores companies selected in a sample.

Table

sample_companies

Fields

Field	Type
id	BIGINT PK
sample_id	BIGINT FK
company_id	BIGINT FK

Field	Type
created_at	DATETIME

Business Rules

- Duplicate companies are not allowed in the same sample.
 - Company identifier validation required during import.
-

16. Survey Activations

Purpose

Stores survey activation history.

Table

survey_activations

Fields

Field	Type
id	BIGINT PK
passage_id	BIGINT FK
activated_by	BIGINT FK
activated_at	DATETIME
notes	TEXT

17. Collection Tracking

Purpose

Tracks participation progress.

Table

collection_tracking

Fields

Field	Type
id	BIGINT PK
passage_id	BIGINT FK
company_id	BIGINT FK
status	ENUM('NOT_STARTED','IN_PROGRESS','COMPLETED')
first_access_at	DATETIME NULL
last_activity_at	DATETIME NULL
submitted_at	DATETIME NULL
created_at	DATETIME

Current Limitation

The update mechanism for these statuses depends on the future integration method with the external questionnaire platform.

18. Notifications

Purpose

Stores all outgoing emails.

Table

notifications

Fields

Field	Type
id	BIGINT PK

Field	Type
user_id	BIGINT FK
channel	ENUM('EMAIL')
type	ENUM('OTP','ACTIVATION','INVITATION','REMINDER','RESET_PASSWORD')
recipient	VARCHAR(255)
status	ENUM('PENDING','SENT','FAILED')
provider_response	TEXT
retry_count	INT DEFAULT 0
sent_at	DATETIME NULL
created_at	DATETIME

19. OTP Codes

Purpose

Stores authentication codes.

Table

otp_codes

Fields

Field	Type
id	BIGINT PK
user_id	BIGINT FK
code	VARCHAR(10)
expires_at	DATETIME
attempts	INT
created_at	DATETIME

20. Login Attempts

Purpose

Stores login history.

Table

login_attempts

Fields

Field	Type
id	BIGINT PK
user_id	BIGINT FK
ip_address	VARCHAR(50)
user_agent	TEXT
success	BOOLEAN
created_at	DATETIME

21. Survey Result Files

Purpose

Stores official INS survey results.

Table

survey_result_files

Fields

Field	Type
id	BIGINT PK
survey_id	BIGINT FK
sector_id	BIGINT FK

Field	Type
file_name	VARCHAR(255)
file_path	TEXT
file_type	ENUM('PDF','EXCEL')
uploaded_by	BIGINT FK
uploaded_at	DATETIME

Business Rules

Only companies belonging to the corresponding sector may access the file.

22. Audit Logs

Purpose

Stores critical actions performed on the platform.

Table

audit_logs

Fields

Field	Type
id	BIGINT PK
user_id	BIGINT FK
action	VARCHAR(255)
entity_type	VARCHAR(100)
entity_id	BIGINT
old_values	JSON
new_values	JSON
ip_address	VARCHAR(50)
created_at	DATETIME

Examples

- Company Created
 - Company Updated
 - Contact Disabled
 - Survey Activated
 - Sample Imported
 - Data Exported
-

23. System Settings

Purpose

Stores configurable parameters.

Table

system_settings

Fields

Field	Type
id	BIGINT PK
setting_key	VARCHAR(100) UNIQUE
setting_value	TEXT
description	TEXT
updated_at	DATETIME

Examples

```
OTP_EXPIRATION_MINUTES = 5
```

```
MAX_LOGIN_ATTEMPTS = 3
```

```
ACCOUNT_LOCK_HOURS = 24
```

```
PASSWORD_ONLY_DURATION_HOURS = 1
```

```
DEFAULT_LANGUAGE = fr
```

```
EMAIL_PROVIDER = brevo
```

24. Entity Relationship Summary

```
roles
  ↓
users
  ↓
contacts
  ↓
companies

surveys
  ↓
survey_assignments

surveys
  ↓
survey_passages
  ↓
survey_questionnaires

survey_passages
  ↓
samples
  ↓
sample_companies
  ↓
companies

survey_passages
  ↓
collection_tracking

surveys
  ↓
survey_result_files

users
  ↓
audit_logs
```

```
users
  ↓
notifications
```

25. Pending Clarifications From INS

The following points remain to be validated before finalizing the database architecture:

1. How the external questionnaire platform communicates questionnaire opening.
2. How questionnaire completion is communicated to the INS platform.
3. How questionnaire responses become accessible to Survey Administrators.
4. Whether an API, webhook, synchronization service, or shared database exists between both systems.

These points may require additional tables and relationships in future versions.

End of Database Specification

Version 2.0 INS Statistical Survey Collection Platform